

SUCCESSFUL CLINICAL AUDIO–METADATA ALIGNMENT DOES NOT IMPLY DISEASE TRANSFER: A PAIRING-CONTROLLED AUDIT

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ABSTRACT

Clinical audio models align recordings with participant metadata, but successful correspondence does not establish transferable disease representation. Metadata mixes disease-related information with demographics, symptoms, history, and cohort structure. We introduce a Pairing-Controlled Transfer Audit comparing correct pairs, metadata shuffled within disease labels, and globally shuffled metadata. On UKCOVID, Correct pairing improved profile retrieval for AST-6L and OPERA-CT and increased sex decodability relative to Within-label (Δ AUROC +0.191 and +0.113), showing that the pairing intervention produced a measurable representational difference. Matched COVID prediction changed by only +0.006 and −0.002. Across a post-hoc UKCOVID HeAR extension and secondary analyses on CODA TB, Cambridge COVID-19 Sounds, and Coswara, retrieval and sex decodability increased consistently, whereas 0/12 dataset–backbone settings established a positive matched-disease gain, and Correct did not consistently outperform frozen raw audio. Correspondence learning, retained participant information, and disease transfer are therefore distinct empirical claims requiring distinct controls.

Index Terms— clinical audio, audio–metadata alignment, contrastive learning, confounding, domain shift

1. INTRODUCTION

Clinical audio models increasingly use participant metadata not only as prediction-time input, but also as supervision for representation learning [1]. Pretrained audio encoders now range from general-audio models such as the Audio Spectrogram Transformer (AST) [2] to respiratory- and health-specific representations such as OPERA and HeAR [3, 4]. Recent multimodal systems further align each recording with clinical text derived from the same participant. RespiraMFM, for example, freezes its audio and text encoders and trains an audio-side projector for audio–clinical-text alignment before downstream multimodal prediction [5].

This setup creates an evaluation problem: a model can become better at matching an audio recording to its associated participant profile without becoming better at predicting disease in a different patient population. A contrastive objective rewards any audio feature that helps identify the assigned metadata profile, while clinical metadata contains demographics, symptoms, medical history, missingness patterns, and cohort structure in addition to disease-related information. Successful alignment can therefore reflect at least three different phenomena: profile correspondence, label-conditioned population association, or disease information that transfers across populations. Alignment loss, profile retrieval, or source-cohort AUROC alone cannot distinguish among them.

The missing control is the *pairing relation itself*. Consider a COVID-positive participant. In our Within-label control, that par-

ticipant’s audio is paired with metadata from a different COVID-positive participant: the disease class is preserved, but the original participant profile is not. Comparing Correct with Within-label therefore asks what the observed audio–profile pairing adds beyond metadata structure already associated with the disease label. Comparing Within-label with Global instead measures the contribution of label-conditioned population association. Disease labels construct these control pairings; they are not included in the metadata text presented to either encoder.

We use these interventions in a *Pairing-Controlled Transfer Audit* that asks three sequential questions: *Was pairing learned? What became more decodable? Does disease prediction improve after population balancing?* Profile retrieval is a manipulation check, information probes characterize the resulting representation, and participant-disjoint matched evaluation tests disease transfer against both Within-label and frozen raw-audio references.

Across UKCOVID and secondary analyses on CODA TB, Cambridge COVID-19 Sounds, and Coswara, we observe a consistent separation between these outcomes. Correct pairing repeatedly improves profile correspondence and, most consistently, sex decodability. In contrast, matched disease effects are small and directionally inconsistent across backbones, and Correct does not consistently outperform the corresponding frozen raw representation. Our claim is not that metadata alignment is generally ineffective. Rather, successful profile alignment and improved disease transfer are separate empirical claims requiring separate controls.

2. DATA AND METHODS

2.1. Datasets and evidence hierarchy

UKCOVID is the discovery dataset [6]. After objective audio filtering, its Standard train contains 20,714 participants. We evaluate Standard test ($n = 11,119$), a covariate-matched test ($n = 1,814$), and a participant-disjoint matched-long sensitivity set ($n = 4,196$). Recruitment source alone predicts COVID with AUROC 0.9966 in Standard train but 0.5000 after matching. This sharp shift is consistent with prior evidence that COVID audio performance can collapse after symptom and demographic adjustment [7]. Because UKCOVID tests informed earlier protocol development, we label it an exploratory discovery audit.

Three external analyses test whether the result shape recurs. CODA TB includes 1,081 eligible participants; we use a frozen 200-participant (100-pair) matched target from its released training data [8]. Cambridge uses 989 participants and a 200-participant target under a predefined strict-COVID reconstruction of its Task-2 cough subset [9]. Coswara uses 1,701 participants and a 200-participant target obtained by a model-blind secondary matching procedure [10]. CODA is a match-first secondary analysis, Cambridge is not an official benchmark replication, and Coswara is a post-hoc stress

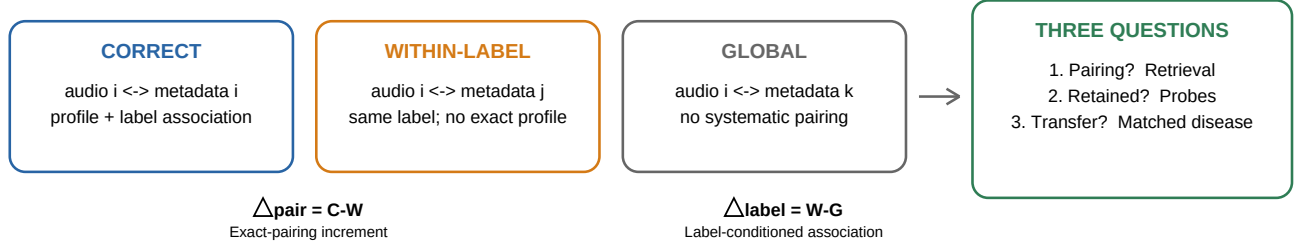


Fig. 1. Pairing-controlled audit. Correct and Within-label preserve the same label-conditioned metadata distribution in expectation, but only Correct preserves exact participant-profile pairing. Δ_{pair} is the exact-pairing increment and Δ_{label} is label-conditioned association.

test. Matching balances prespecified measured covariates; it does not eliminate unmeasured confounding.

2.2. Audio representations and pairing interventions

We use two frozen audio backbones for the primary UKCOVID audit, chosen to represent complementary pretraining regimes. **AST-6L** retains the first six Transformer blocks of the 12-layer AudioSet-finetuned AST encoder [2]. This depth was fixed before the present audit from an independent respiratory-audio comparison: six blocks outperformed the full 12-layer model on both ICBHI wheeze and crackle detection while using about half the parameters, and the same comparison favored AST-6L on three of four patient-level KAUH tasks. It was not selected using UKCOVID transfer performance.

OPERA-CT is the released frozen 768-dimensional contrastive Transformer from the OPERA respiratory acoustic foundation-model family [3]. Because its pretraining corpus includes UKCOVID training audio, we treat it as a domain-pretrained comparator rather than an independent pretraining condition; the present UKCOVID targets remain participant-disjoint from our training split. **HeAR** is added post hoc on UKCOVID and in the external analyses as a third frozen health-acoustic representation [4]; it is not promoted to a primary UKCOVID backbone.

Recordings use each backbone’s frozen preprocessing pipeline and are aggregated to one participant-level audio representation before alignment. A frozen Phi-2 encoder [11] represents a deterministic metadata schema containing available demographics, smoking status, respiratory history, symptoms, and explicit missing values. Disease labels, participant identifiers, recruitment source, country/site, platform, device, timestamps, and recording artefacts are excluded from the alignment text.

For participant i , let a_i , m_i , and y_i denote audio, metadata, and disease label. The three training arms are

$$\begin{aligned} \text{Correct: } & a_i \leftrightarrow m_i, \\ \text{Within-label: } & a_i \leftrightarrow m_j, \quad y_i = y_j, \quad i \neq j, \\ \text{Global: } & a_i \leftrightarrow m_k, \quad i \neq k. \end{aligned} \quad (1)$$

Global removes systematic participant- and label-conditioned pairing, although accidental same-label or duplicate-profile matches can remain. Within-label breaks the observed participant–profile relationship while preserving label-conditioned metadata structure in expectation. We denote Correct-minus-Within as $\Delta_{\text{pair}} = C - W$, the increment associated with observed participant–profile pairing beyond label-conditioned association. We denote Within-minus-Global as $\Delta_{\text{label}} = W - G$, which measures label-conditioned population association. Neither is a causal estimand, and Δ_{pair} is

not a participant-identity effect. $C - R$ compares Correct with the frozen raw audio representation.

Only an audio-side projector is trained, using the one-directional audio-to-text InfoNCE objective from the audited Stage-1 implementation:

$$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(\bar{f}_\theta(a_i)^\top \bar{t}_i / \tau)}{\sum_j \exp(\bar{f}_\theta(a_i)^\top \bar{t}_j / \tau)}, \quad \tau = 0.07, \quad (2)$$

where bars denote ℓ_2 normalization and t_i is frozen text embedding. Each arm uses five seeds and 500 epochs; within a seed, arms share initialization, audio batches, and dropout streams, differing only in pairing. Pair assignments remain fixed during training, and only the final checkpoint is evaluated. Matched data select no epoch, projector, readout, or hyperparameter. This is a paper-faithful Stage-1 reimplement, not a reproduction of the full downstream Phi-2/LoRA system.

2.3. Audit endpoints and statistics

Correspondence: audio queries retrieve metadata profiles on source validation. Because multiple participants can share a schema, identical profiles count as the same semantic target; we report macro-profile MRR so frequent schemas do not dominate. **Information:** identical regularized probes decode sex, age, symptoms, acquisition/cohort variables, and disease. Decodability establishes information availability, not causal classifier use. **Transfer:** source data alone select and calibrate downstream readouts; paired AUROC, NLL, and Brier score are then evaluated once on matched targets. Intervals use participant-by-seed hierarchical bootstrap, retaining paired predictions.

Frozen raw-audio and metadata-only references check whether alignment improves over simpler representations. A fixed MLP checks linear-readout limitation. Finally, a one-dimensional calibrator fitted on participant-disjoint matched-long and transported to matched separates confidence transport from ranking.

3. RESULTS

3.1. Correct pairing improves correspondence and sex decodability

On UKCOVID validation, Correct improved macro-profile MRR over Within-label for both backbones. Δ_{pair} was +0.00318 [0.00161, 0.00492] for AST and +0.00323 [0.00155, 0.00500] for OPERA. All five seed-level effects were positive. Retrieval is a manipulation check: later disease results cannot be explained simply by failure to learn the assigned pairing.

The strongest probe effect was sex. On UKCOVID matched participants, Δ_{pair} sex-probe ΔAUROC was +0.1912 [0.1669, 0.2151] for AST and +0.1125 [0.0943, 0.1311] for OPERA. The post-hoc HeAR extension gave retrieval ΔMRR +0.00711 [0.00476, 0.00971] and sex ΔAUROC +0.0862 [0.0734, 0.0995]. Age increased for AST but not OPERA; acquisition and cohort effects were less consistent. Raw audio generally had higher absolute demographic decodability than aligned representations, so alignment did not create demographic information absent from audio; it selectively retained it relative to Within-label.

3.2. Correspondence gains lack consistent disease-transfer gains

On UKCOVID matched, disease Δ_{pair} ΔAUROC was +0.0062 [-0.0144, 0.0279] for AST and -0.0020 [-0.0191, 0.0168] for OPERA. For AST, raw, Correct, and Within AUROC were 0.538, 0.529, and 0.523; for OPERA they were 0.577, 0.555, and 0.557. A small AST fusion effect did not replicate with OPERA, and Correct fusion did not outperform metadata plus raw audio. Post-hoc HeAR raw, Correct, and Within AUROC were 0.542, 0.545, and 0.543, giving Δ_{pair} +0.0029 [-0.0115, 0.0165]. A fixed MLP also failed to reveal a hidden effect: audio-only Δ_{pair} was +0.0050 [-0.0154, 0.0270] for AST and -0.0052 [-0.0208, 0.0110] for OPERA.

Figure 2 summarizes the result shape. Across all 12 dataset-backbone settings, retrieval point estimates were positive (11/12 intervals excluded zero) and sex Δ_{pair} was positive with 12/12 intervals excluding zero. In contrast, matched disease Δ_{pair} produced six positive and six negative point estimates, and 0/12 intervals established a positive gain. This count is descriptive, not a meta-analysis: targets contain different diseases, MRR libraries differ, and external analyses have different evidence status. Correct likewise showed no consistent advantage over frozen raw audio.

3.3. Label association, fusion, and calibration

Within-label and Global also separated in the source cohort. For UKCOVID AST, source Δ_{label} probe ΔAUROC was +0.0646 for COVID, +0.0758 for cough, and +0.0647 for recruitment source; all three intervals excluded zero. On matched participants, the COVID increment fell to +0.0126 with an interval crossing zero, while cough remained positive (+0.0803). Thus, label-conditioned metadata association is itself learnable and would be conflated with exact pairing by a Correct-versus-Global comparison alone.

Direct fusion did not provide a robust alternative positive result. On UKCOVID matched, adding raw audio to metadata changed AUROC by only +0.0028 [-0.0016, 0.0073] for AST and +0.0041 [-0.0006, 0.0087] for OPERA. Metadata plus Correct exceeded metadata plus Within for AST (+0.0156 [0.0034, 0.0276]) but not OPERA (+0.0021 [-0.0068, 0.0107]); neither backbone’s Correct fusion beat metadata plus raw audio. This isolates a small, backbone-specific ranking signal rather than a reproducible alignment advantage.

With source calibration, UKCOVID Correct had worse matched NLL than Within, although AUROC differences were negligible. The change in negative NLL was -0.0249 [-0.0443, -0.0069] for AST and -0.0176 [-0.0335, -0.0010] for OPERA. Target calibration reduced both gaps to approximately zero without changing AUROC. The source NLL penalty therefore primarily reflects confidence unsupported after cohort shift, not demonstrated destruction of disease ranking; recalibration repairs probability scale but creates no transferable discrimination.

4. DISCUSSION

The audit separates two outcomes that conventional evaluation can conflate. Correct audio-metadata alignment repeatedly improved retrieval of participant profiles, showing that the pairing intervention produced a measurable correspondence signal. Yet the most reproducible additional channel was participant sex, while disease improvements were small, directionally inconsistent, and not robust to raw-audio controls or readout changes. In plain terms, the model became better at matching a recording to the kind of participant described by a metadata card, but not reliably better at diagnosing disease after measured patient composition was balanced.

Within-label shuffling is essential to this conclusion. Correct versus Global alone would mix exact profile correspondence with label-conditioned population association. Retrieval then verifies the manipulation, probes describe the retained channels, and matched evaluation asks the clinical transfer question. Future clinical audio-metadata studies should minimally report these controls together with a frozen raw reference and calibration under population shift.

The same correspondence-transfer separation appeared with a general-audio AST representation and a respiratory-domain OPERA representation, reducing the likelihood that the result is specific to one encoder family.

This changes how positive alignment results should be interpreted. Low alignment loss or high profile retrieval establishes that paired supervision was absorbed; it does not say which part of a heterogeneous profile was used. A source-domain disease gain can then arise from label-conditioned patient composition, particularly when recruitment and label are coupled. Only a prespecified participant-disjoint target that balances measured patient context asks whether the gain survives the intended population change. Pairing is therefore a manipulation check, probes are representation diagnostics, and matched disease prediction is the transfer endpoint.

The recurring sex result also requires careful wording. A probe shows that sex is decodable, not that the disease classifier causally relies on it. Moreover, raw audio often contains more demographic information than any aligned representation. Correct therefore appears to preserve this channel relative to a Within-label projector rather than create a new demographic attribute. This distinction prevents a large Δ_{pair} probe effect from being misread as participant identification or newly introduced bias.

The evidence remains bounded. UKCOVID is a discovery audit; CODA, Cambridge, and Coswara are secondary or post-hoc sensitivities with 100-pair targets and wide intervals. Matching balances measured covariates only, probes do not establish causal use, and the experiment audits one Stage-1-style projector rather than a complete multimodal reasoning system. Our synthetic study also failed its pre-registered general-mechanism gates, so these recurring patterns are not presented as a universal causal law. A proposed disease-invariant repair was not trained because the available single-dataset cohorts lacked the prespecified support for cross-environment same-label positives and within-environment opposite-label negatives. This is a data-feasibility limitation, not a negative result for the repair method.

5. CONCLUSION

Correct pairing reproducibly improved participant-profile correspondence, especially sex, but did not yield a robust cross-backbone improvement in covariate-balanced disease prediction or consistently outperform raw audio. Successful audio-metadata correspondence is therefore not sufficient evidence of a transferable disease

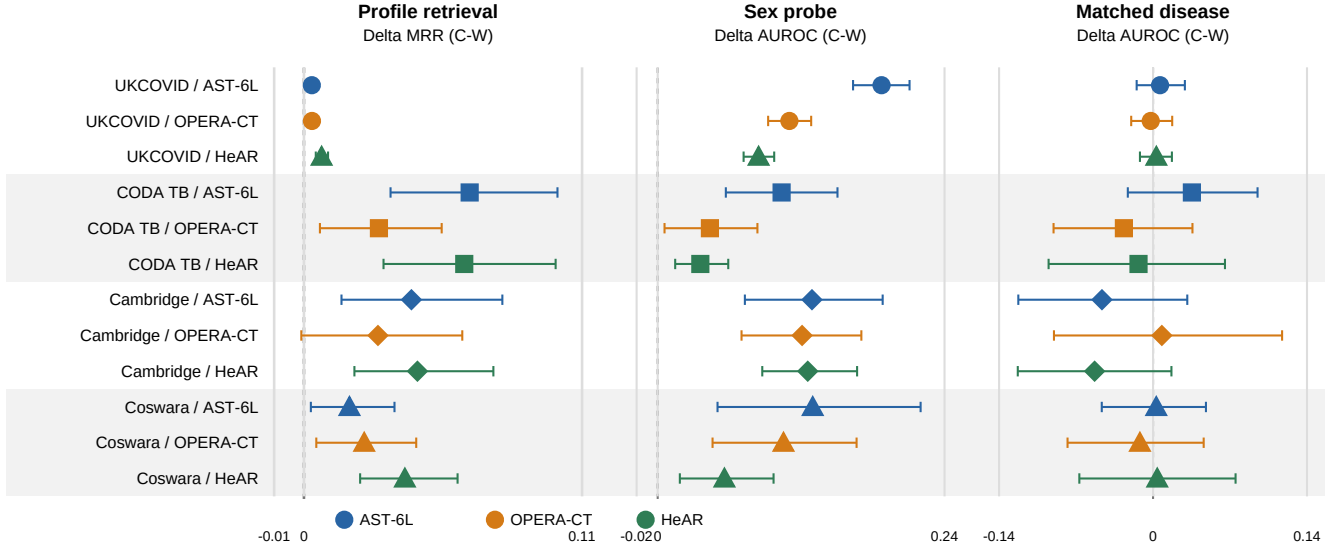


Fig. 2. Δ_{pair} estimates with 95% paired bootstrap intervals. The 12 settings comprise UKCOVID (primary: AST, OPERA; post-hoc: HeAR) and three external sensitivities (AST, OPERA, HeAR each). Retrieval CIs excluded zero in 11/12 settings and sex-probe CIs in 12/12, whereas no matched disease CI established a positive gain. Panel scales differ; estimates are not pooled because profile libraries, diseases, and evidence status differ.

representation; correspondence, retained information, and disease transfer must be evaluated separately.

Compliance with Ethical Standards

This work is a secondary analysis of de-identified datasets accessed and used under their respective data-use terms. No new participants were recruited and no clinical decisions were made from model outputs.

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